

# Communicating with Data – Visual Encoding 2

Dr. Ab Mosca (they/them)

# Plan for Today

- Recap:
  - Visualization building blocks
  - Mapping data to visuals
- More complex examples
- Practice

# Recap

Find a small group to work with. Answer the following:

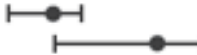
1. What 3 types of marks that make up visualizations?
2. What is the difference between a magnitude and an identity visual channel?
3. Name 6 magnitude visual channels.
4. Name 3 identity visual channels.

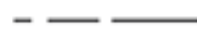
- Remember... **Big idea behind visualization**
  - Visual Encoding:** Map data dimensions to visual dimensions in a principled way

## Recap

### ➔ **Magnitude Channels: Ordered Attributes**

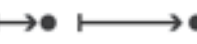
Position on common scale 

Position on unaligned scale 

Length (1D size) 

Tilt/angle 

Area (2D size) 

Depth (3D position) 

Color luminance 

Color saturation 

Curvature 

Volume (3D size) 

### ➔ **Identity Channels: Categorical Attributes**

Spatial region 

Color hue 

Motion 

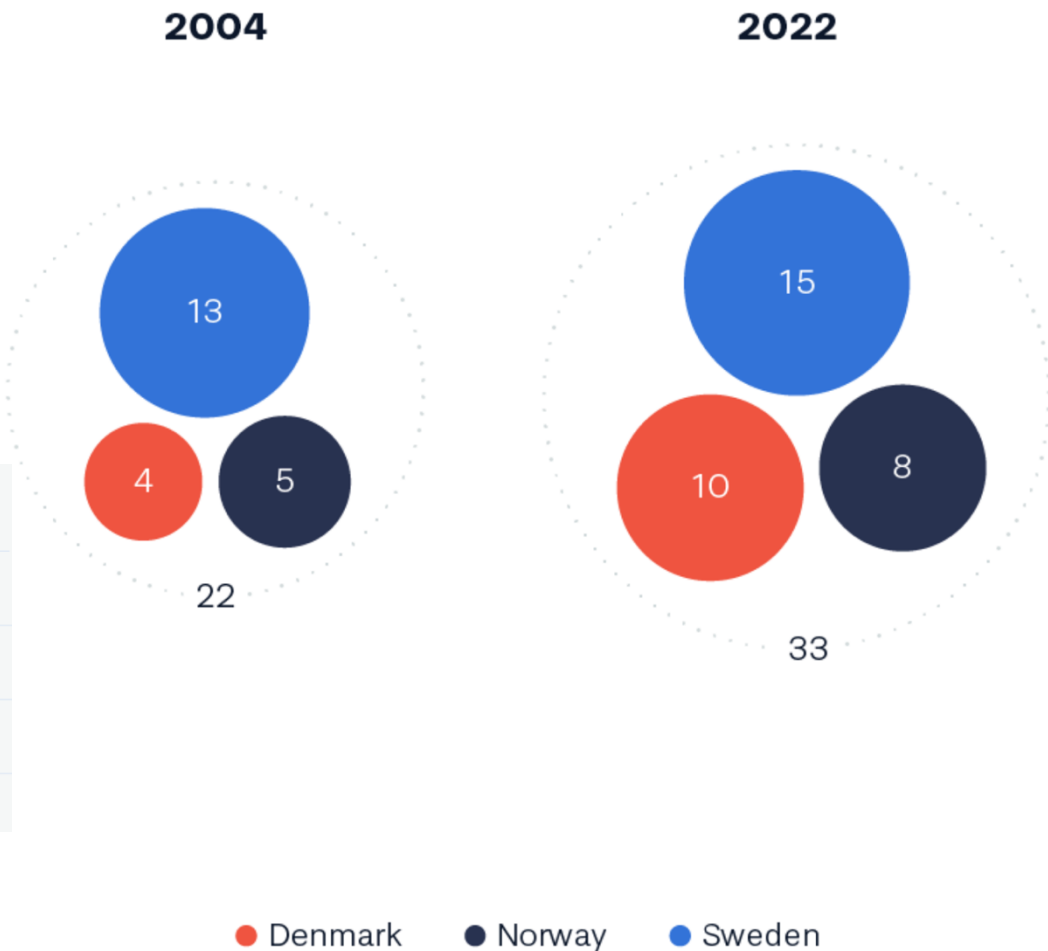
Shape 

# Recap

Find a small group to work with. Answer the following:

1. What marks and channels compose this vis? What is the visual encoding?

| Number of World Heritage Sites |        |         |        |
|--------------------------------|--------|---------|--------|
|                                | Norway | Denmark | Sweden |
| 2004                           | 5      | 4       | 13     |
| 2022                           | 8      | 10      | 15     |

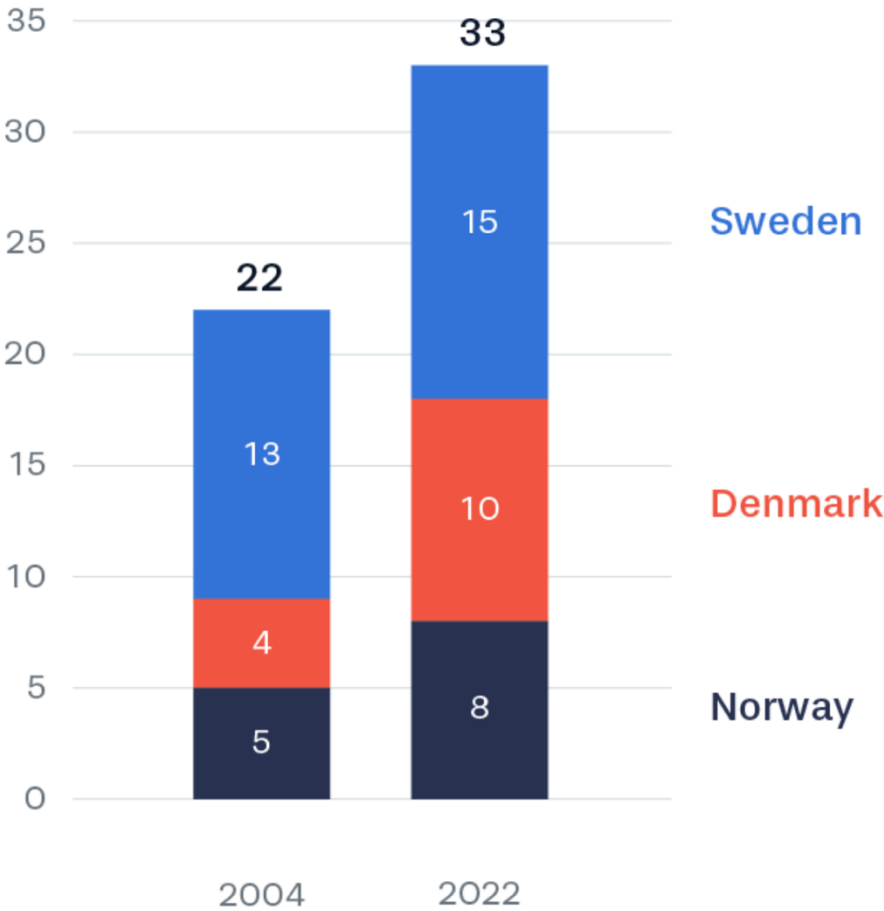


# Recap

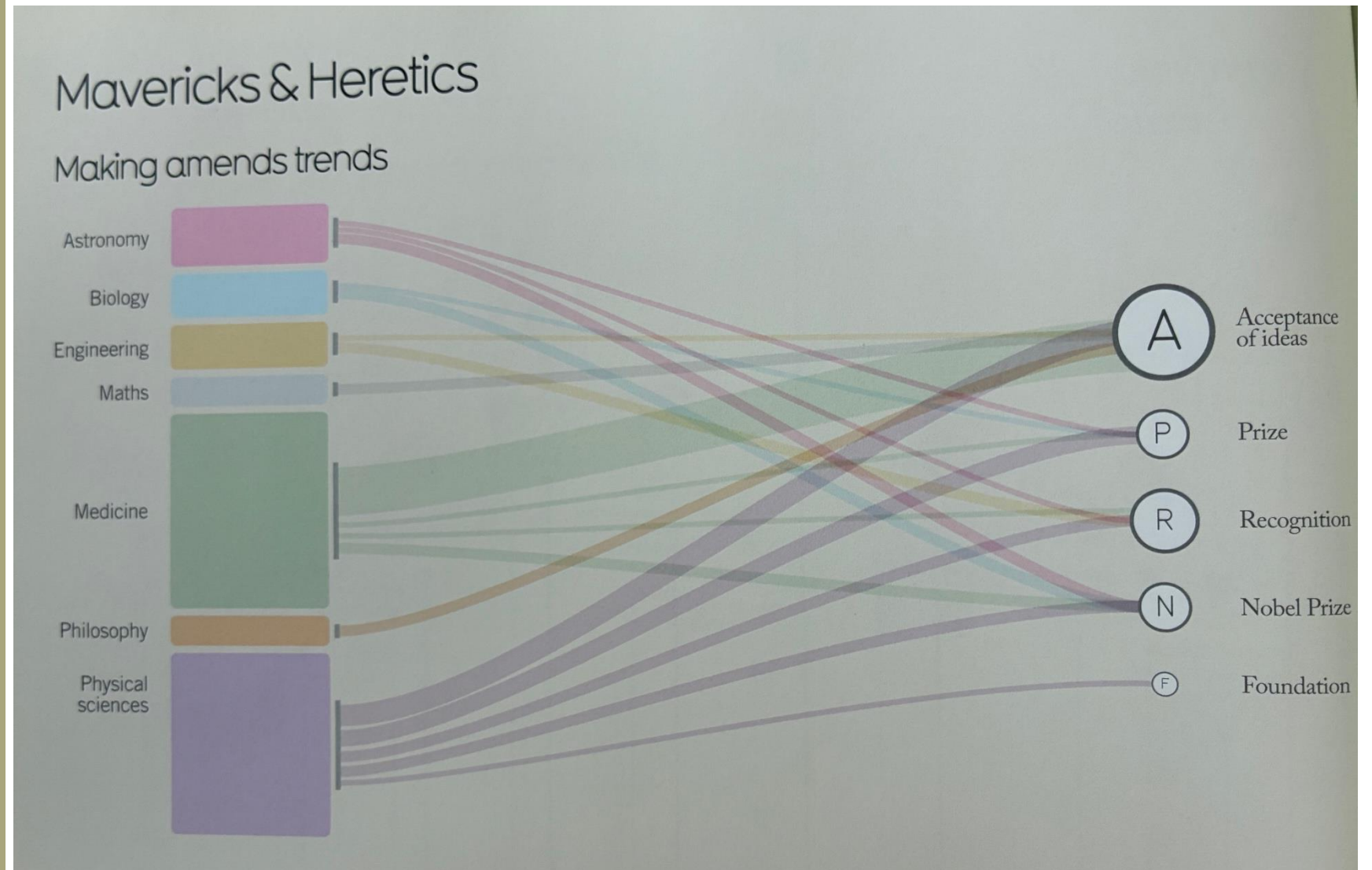
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# Visual Encoding Examples

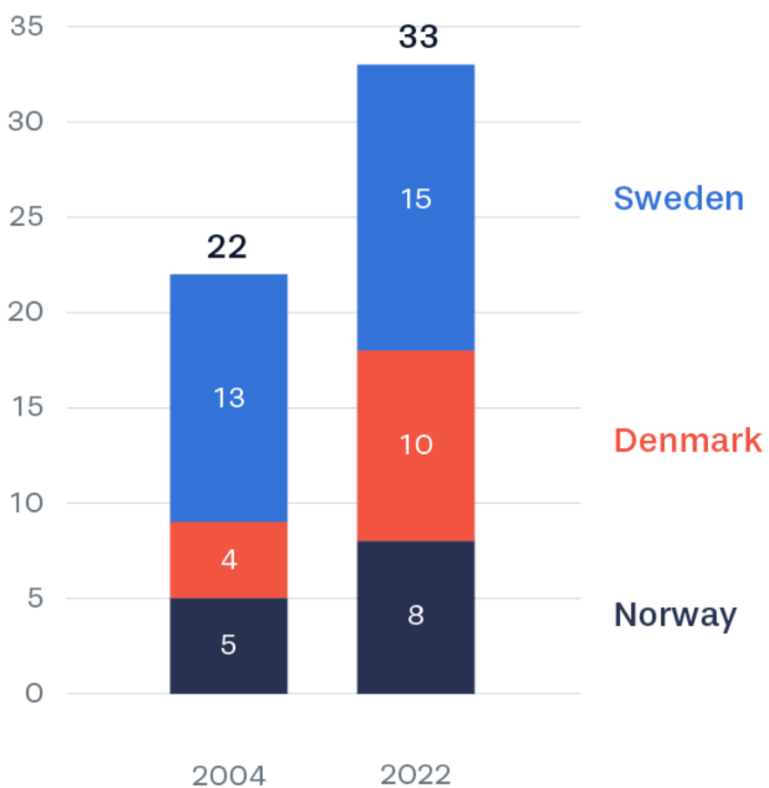
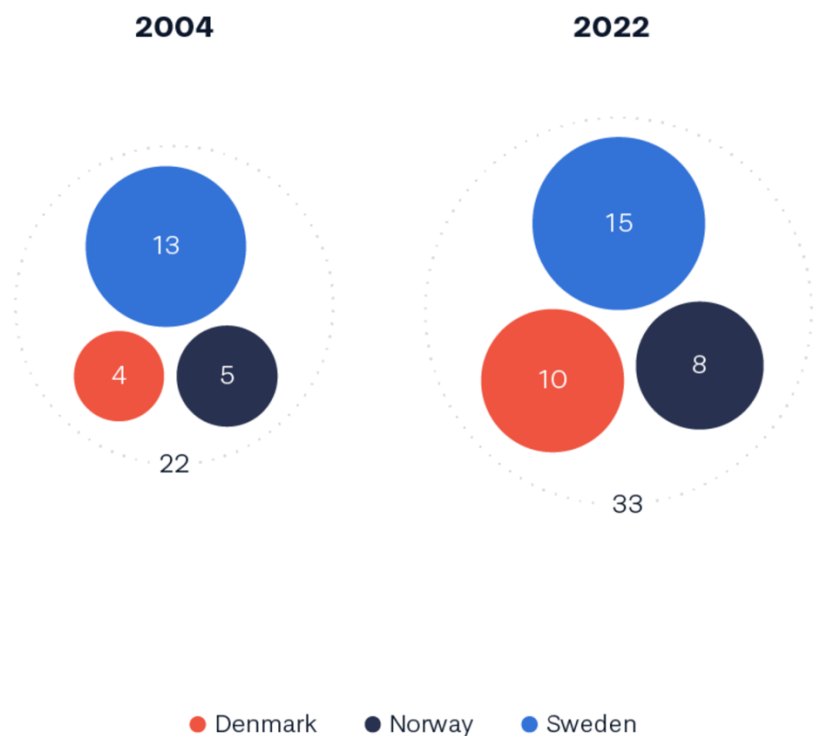


McCandless, D. (2014). *Knowledge is beautiful: Impossible ideas, invisible patterns, hidden connections—visualized*. HarperCollins.

# Recap

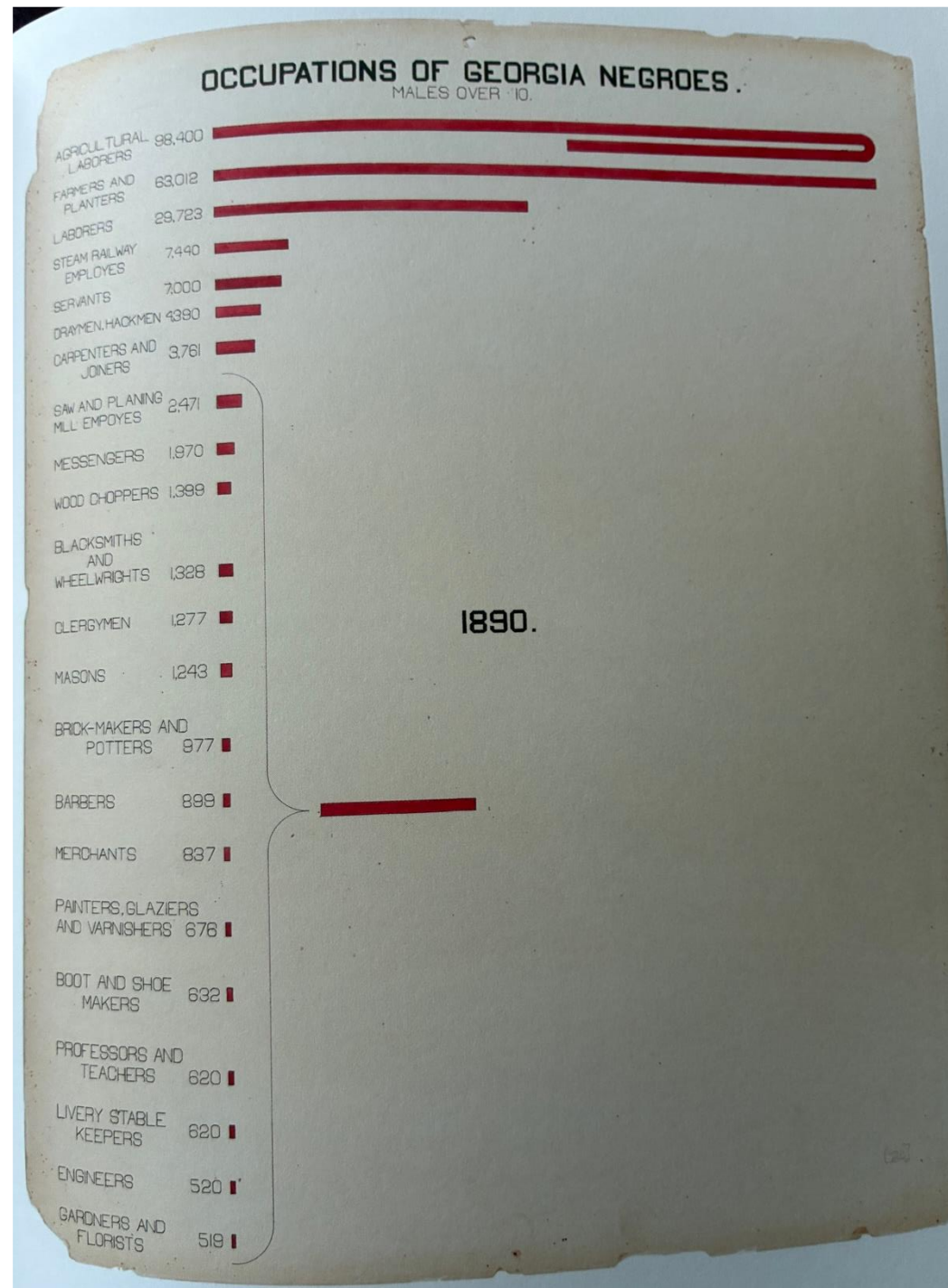
Find a small group to work with. Answer the following:

1. Which is better? Why?





# Visual Encoding Examples

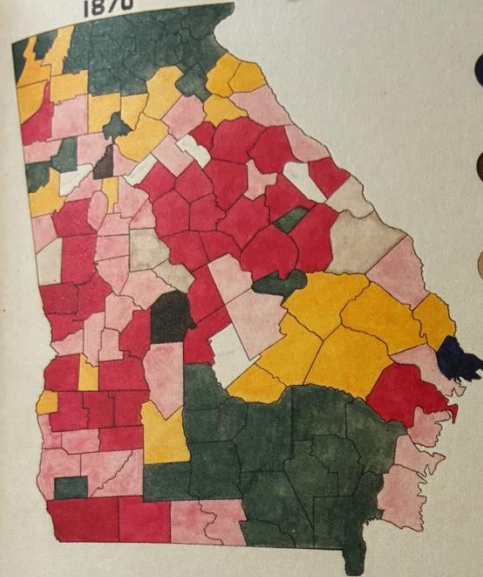


Battle-Baptiste, W., & Rusert, B. (Eds.). (2018).  
*W. E. B. Du Bois's data portraits: Visualizing  
Black America*. Princeton Architectural Press.

# Visual Encoding Examples

## NEGRO POPULATION OF GEORGIA BY COUNTIES.

1870



● BETWEEN 20,000 AND 30,000

● 15,000 TO 20,000

● 10,000 TO 15,000

1880



● 5,000 TO 10,000

● 2,500 TO 5,000

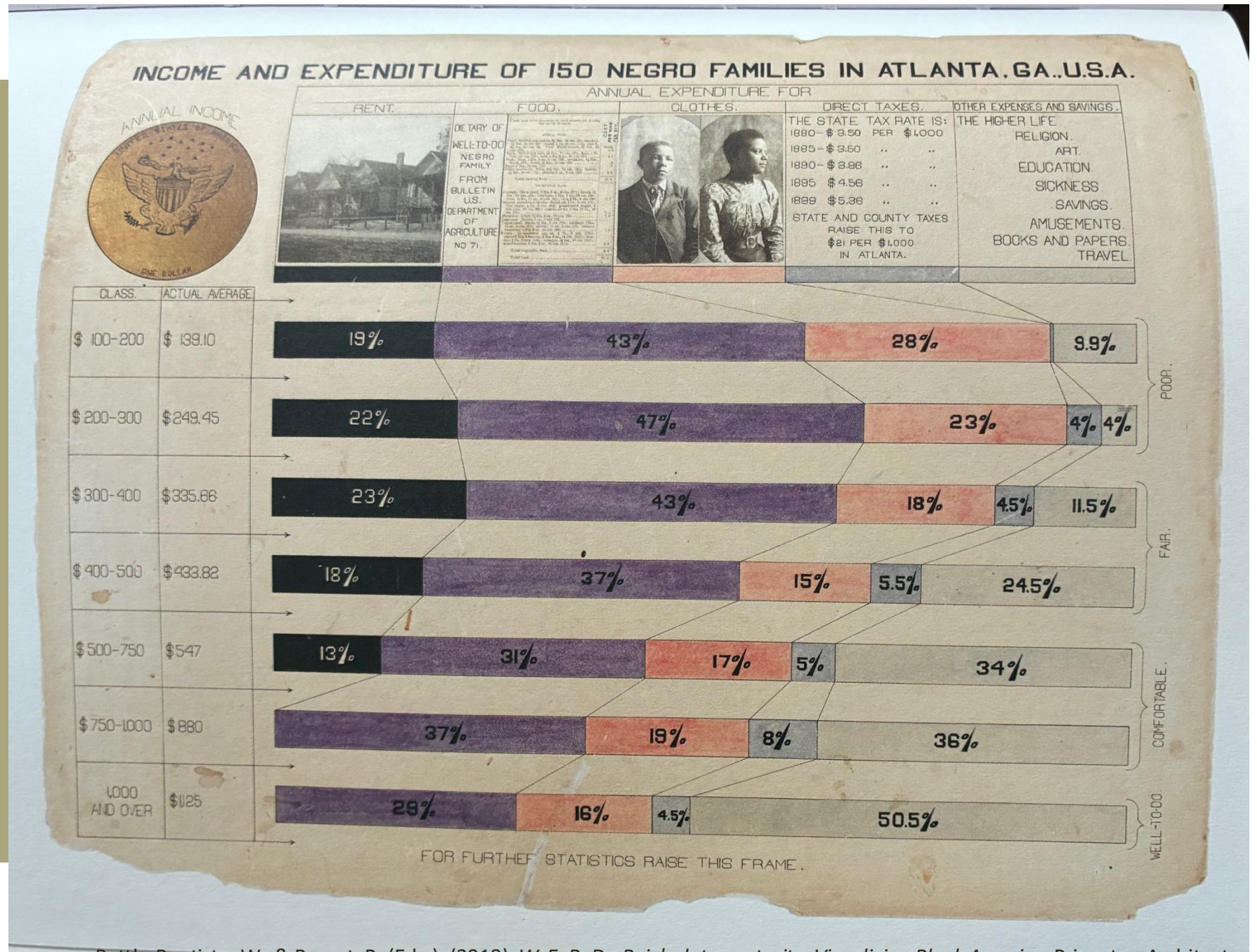
● 1,000 TO 2,500

● UNDER 1,000

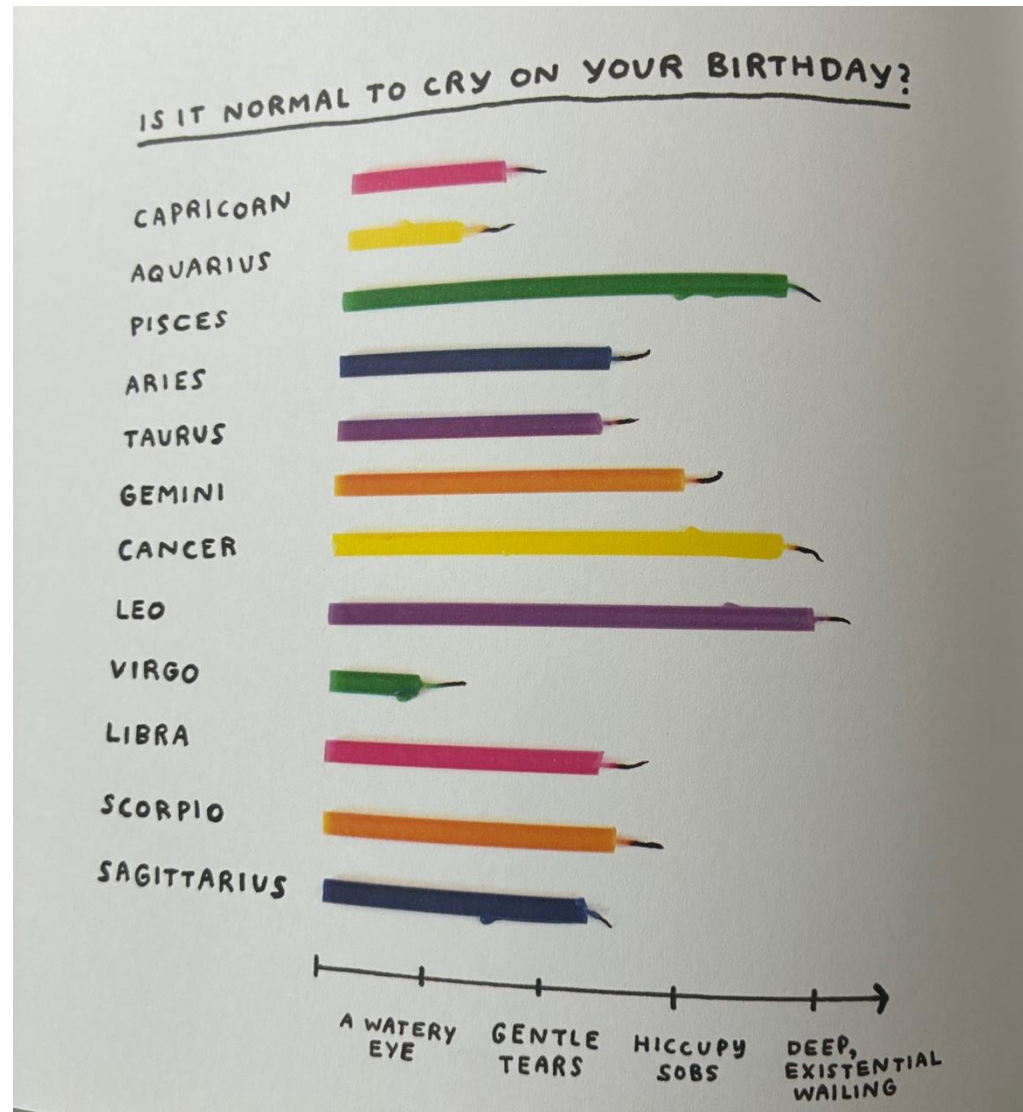
Battle-Baptiste, W., & Rusert, B. (Eds.). (2018).  
*W. E. B. Du Bois's data portraits: Visualizing  
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# Visual Encoding Examples



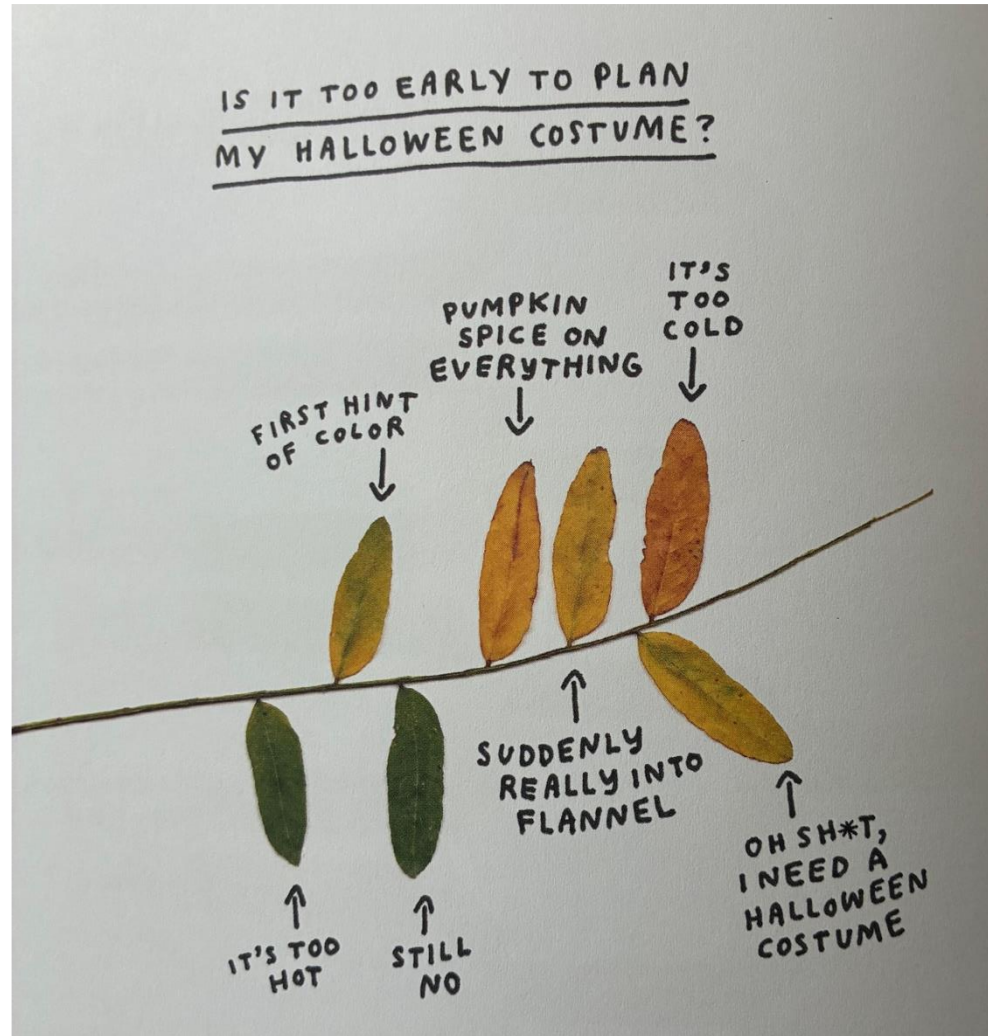
# Visual Encoding Examples



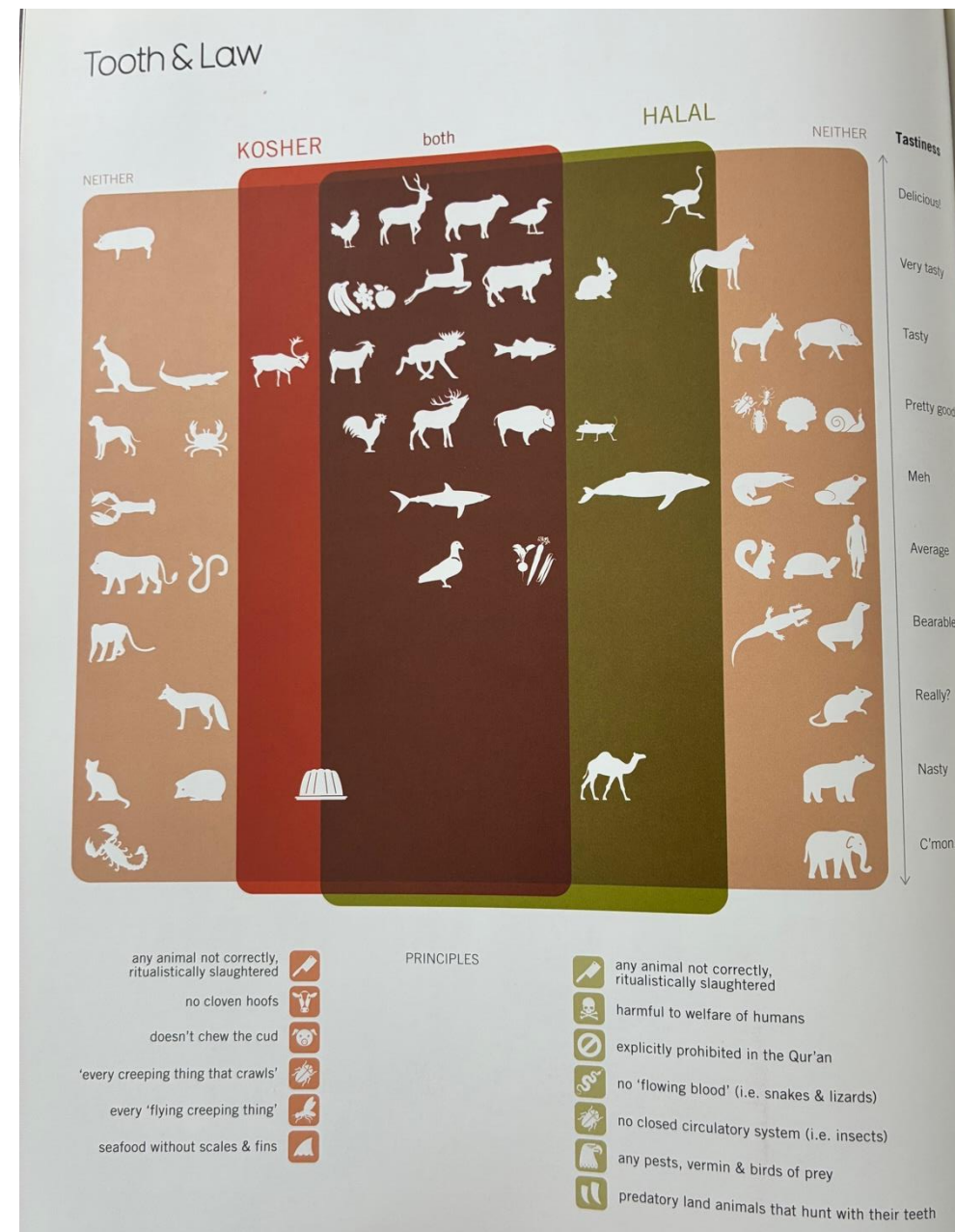
Rial, M. (2019). *Am I overthinking this? Over-answering life's questions in 101 charts*



# Visual Encoding Examples

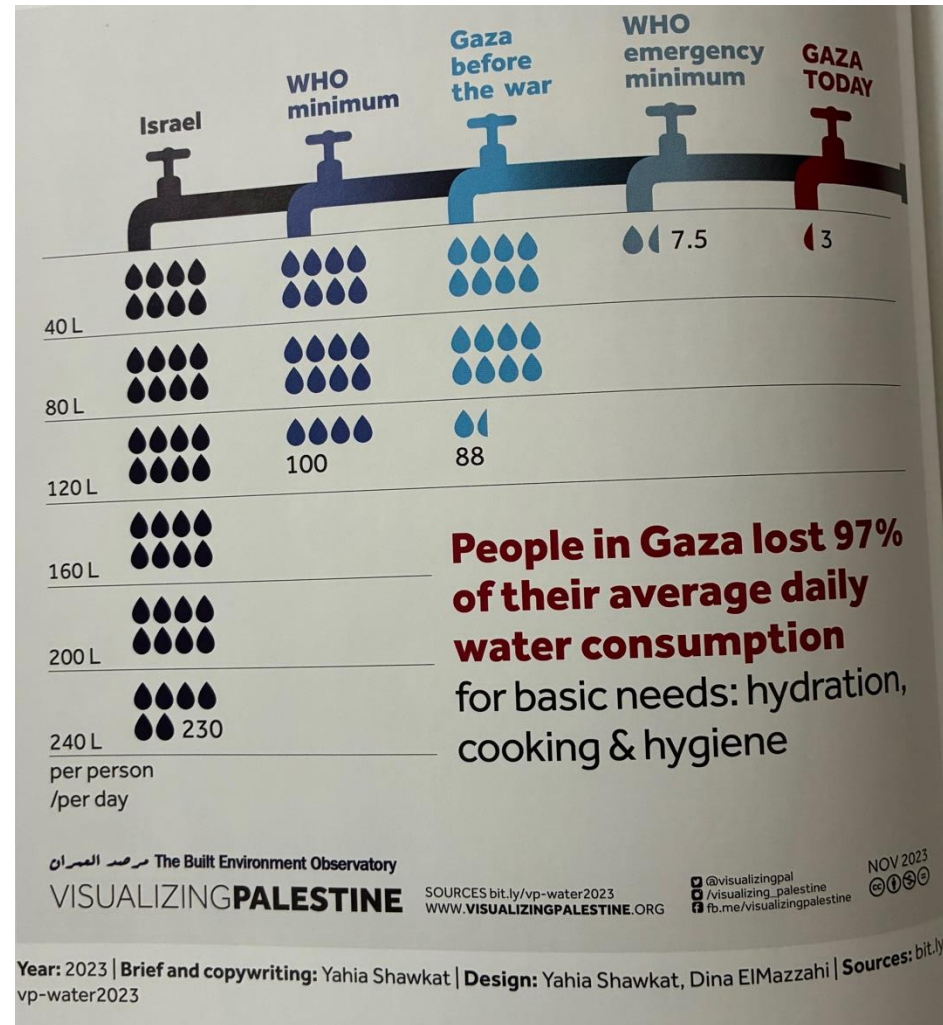


# Visual Encoding Examples



McCandless, D. (2014). *Knowledge is beautiful: Impossible ideas, invisible patterns, hidden connections—visualized*. HarperCollins.

# Visual Encoding Examples



Anderson, J., Batarseh, A., & El Gazzar, Y. (Eds.). (2024). *Visualizing Palestine: A Chronicle of Colonialism and the Struggle for Liberation*. Haymarket Books



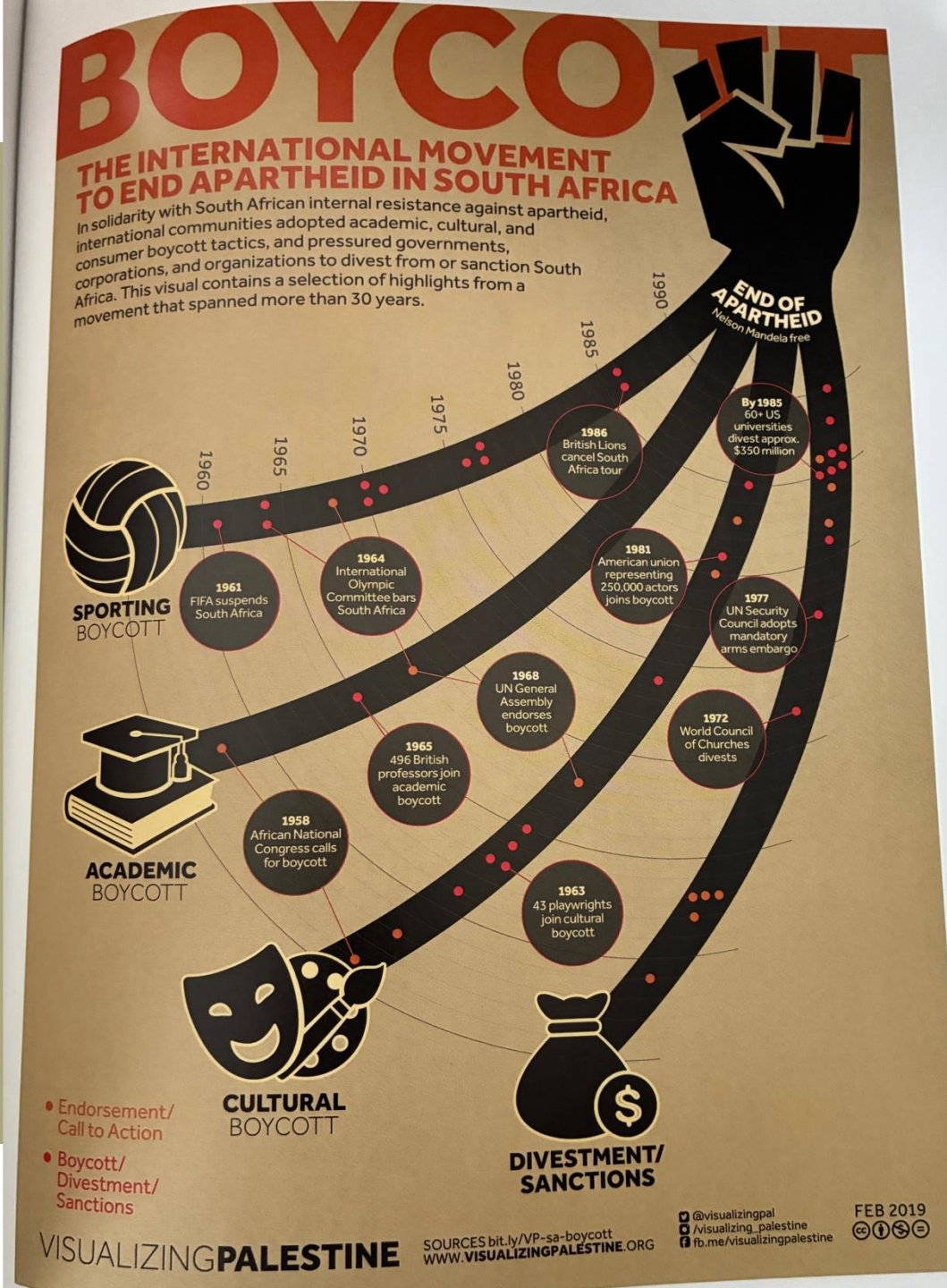
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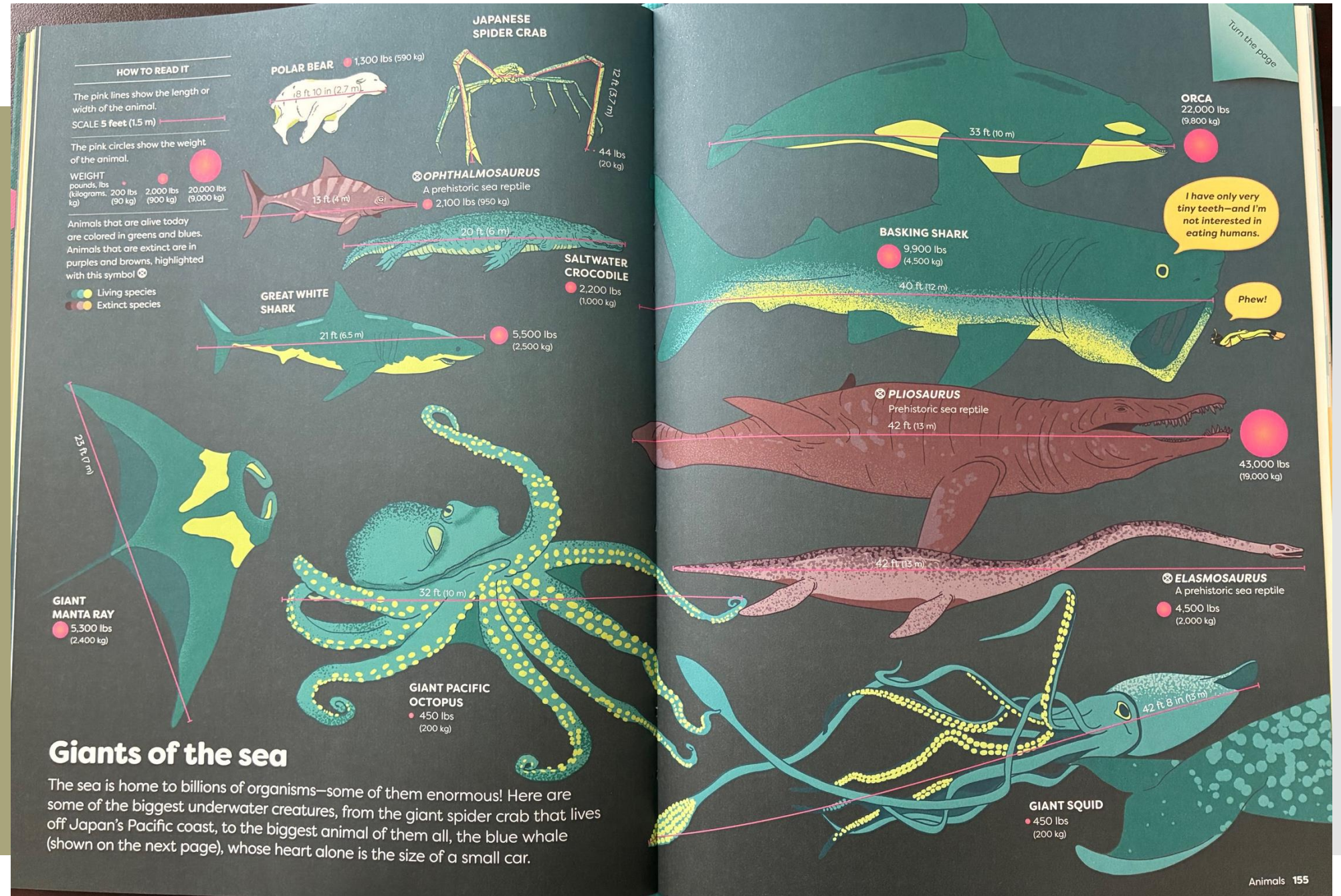
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# Visual Encoding Examples





# Visual Encoding Examples

## The strongest animals on Earth

Which animal is stronger, an elephant or an ant? The elephant can carry the heavier weight, of course. But perhaps a fairer way to decide is to measure how much weight each animal can carry relative to the weight of its own body. Looking at it this way, small but powerful animals, such as ants and beetles, are even stronger—relative to the size of their bodies—than gorillas, bears, and elephants!



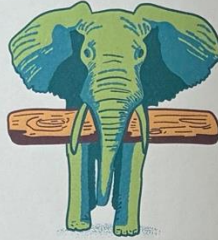
### HARPY EAGLE

The harpy eagle can carry 17 lbs (7.5 kg), the same as **(1 time)** its own weight.



### GRIZZLY BEAR

A grizzly bear can carry 550 lbs (250 kg), the same as **(1 time)** its own weight.



### AFRICAN ELEPHANT

This elephant can carry 20,000 lbs (9,000 kg), **1.5 times** its own weight of 13,000 lbs (6,000 kg).



### TIGER

A tiger can carry 880 lbs (400 kg), twice **(2 times)** its own weight of 440 lbs (200 kg).



### MUSK OX

A musk ox can carry 1,250 lbs (570 kg), twice **(2 times)** its own weight of 630 lbs (285 kg).

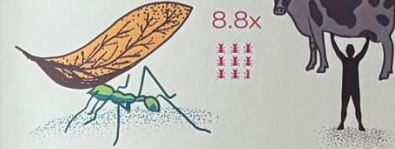
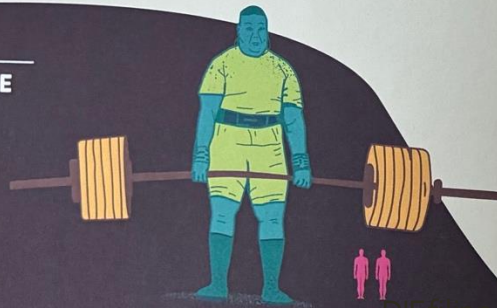


### GORILLA

A gorilla can carry 1,400 lbs (640 kg), **4 times** its own weight of 350 lbs (160 kg).

### HOW STRONG ARE HUMANS?

The world record-holding weightlifter **Tamara Walcott** can lift 641 lbs (291 kg), twice **(2 times)** her own weight of 295 lbs (134 kg).



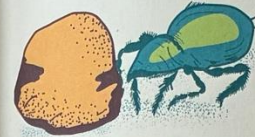
### LEAFCUTTER ANT

This remarkable ant can carry **nearly 9 times** its own weight of 0.0007 oz (0.02 g). This is equivalent to a **human carrying a cow!**



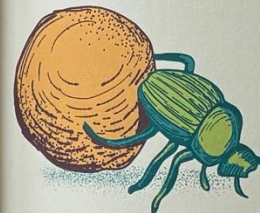
### RHINOCEROS BEETLE

A rhinoceros beetle can carry **100 times** its own weight of 0.08 oz (2.4 g). This is equivalent to a **human carrying an elephant!**



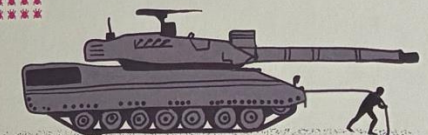
### ORIBATID MITE

This mite can pull **530 times** its own weight of 0.0000035 oz (0.0001 g). This is equivalent to a **human pulling the weight of a Brachiosaurus dinosaur!**



### DUNG BEETLE

A dung beetle can pull **more than 1,100 times** its own weight of 0.003 oz (0.09 g). This is equivalent to a **human pulling a tank!**





# Visual Encoding Examples

## Which animals kill the most humans?

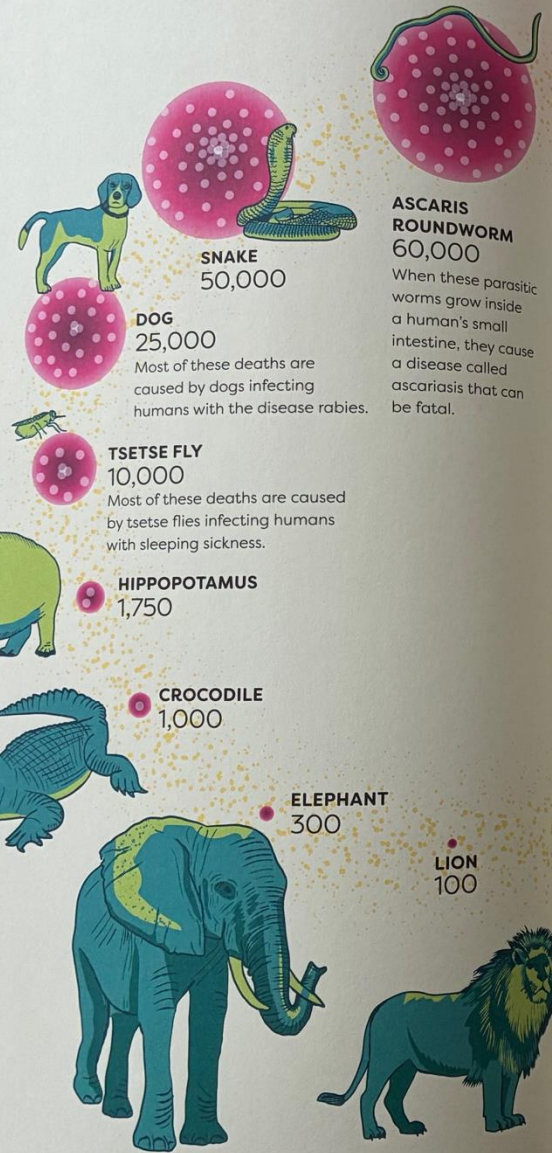
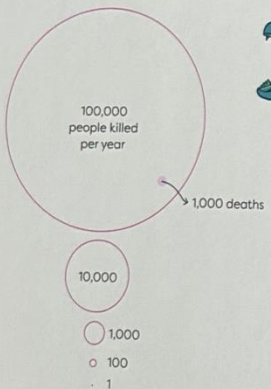
Which animal do you think poses the greatest danger to humans? If you said a hungry shark or a poisonous spider, you would be... completely wrong! Because the deadliest animal for humans is the tiny mosquito.

### WHY ARE MOSQUITOES SO DEADLY?

Mosquitoes do not kill humans directly. But they do transmit dangerous diseases between humans and other animals when they bite them and drink their blood. The deadliest of these diseases is malaria, which is found only in certain parts of the world but has killed millions of people throughout human history.

#### HOW TO READ IT

The size of the pink circles shows the estimated number of humans that each animal species kills per year.



#### ASCARIS ROUNDWORM

60,000  
When these parasitic worms grow inside a human's small intestine, they cause a disease called ascariasis that can be fatal.

#### FRESHWATER SNAIL

200,000  
Freshwater snails carry parasitic worms that can infect humans with a potentially fatal disease called schistosomiasis, which is also known as "snail fever." Schistosomiasis infects nearly 250 million people around the world each year.

Mosquitoes like me can transmit as many as 15 diseases, including malaria, yellow fever, and Zika.

#### MOSQUITO

725,000  
These deaths are mostly caused by mosquitoes infecting humans with the disease malaria.

#### SHARK

5  
On average, sharks kill just five humans per year. Humans, however, kill more than 100 million sharks each year for their fins or teeth, or for sport.

D'Elipippo, Valentina, Andrew Pettie, and Conrad Quilty-Harper. *Britannica's Encyclopedia Infographica: 1,000s of Facts & Figures About Earth, Space, Animals, the Body, Technology, & More Revealed in Pictures*. Britannica Books, 2023.



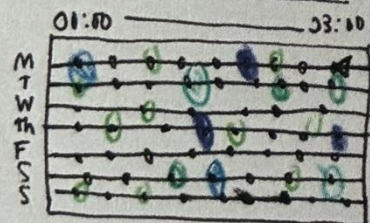
# DEAR DATA - WEEK 12

## A WEEK OF PEOPLE

THE DATA: I TRACKED THE NUMBER OF PEOPLE IN MY VICINITY (SO THE # OF PEOPLE IN THE ROOM I WAS IN, OR THE # OF PEOPLE IN ANY ENCLOSED SPACE). THIS WAS HARD TO GATHER, HENCE ME RESORTING TO REPRESENTING THE DATA THROUGH USING BANDING. NOTE: SOME DATA IS MISSING DUE TO MY HAPHAZARD DATA GATHERING!

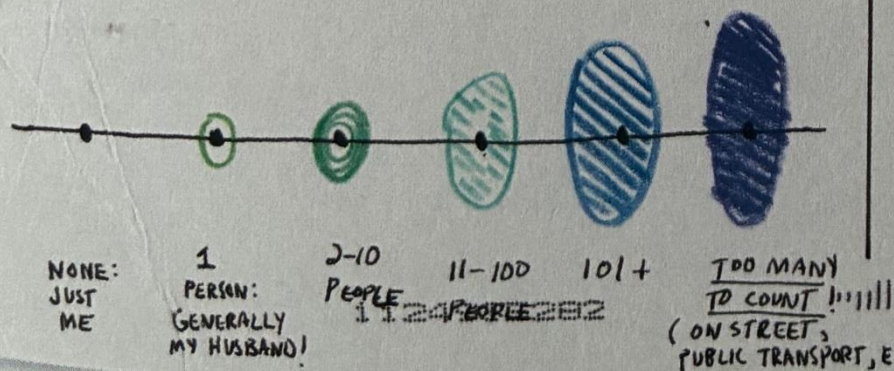
### HOW TO READ IT:

(NOT MUCH SLEEP THIS WEEK)

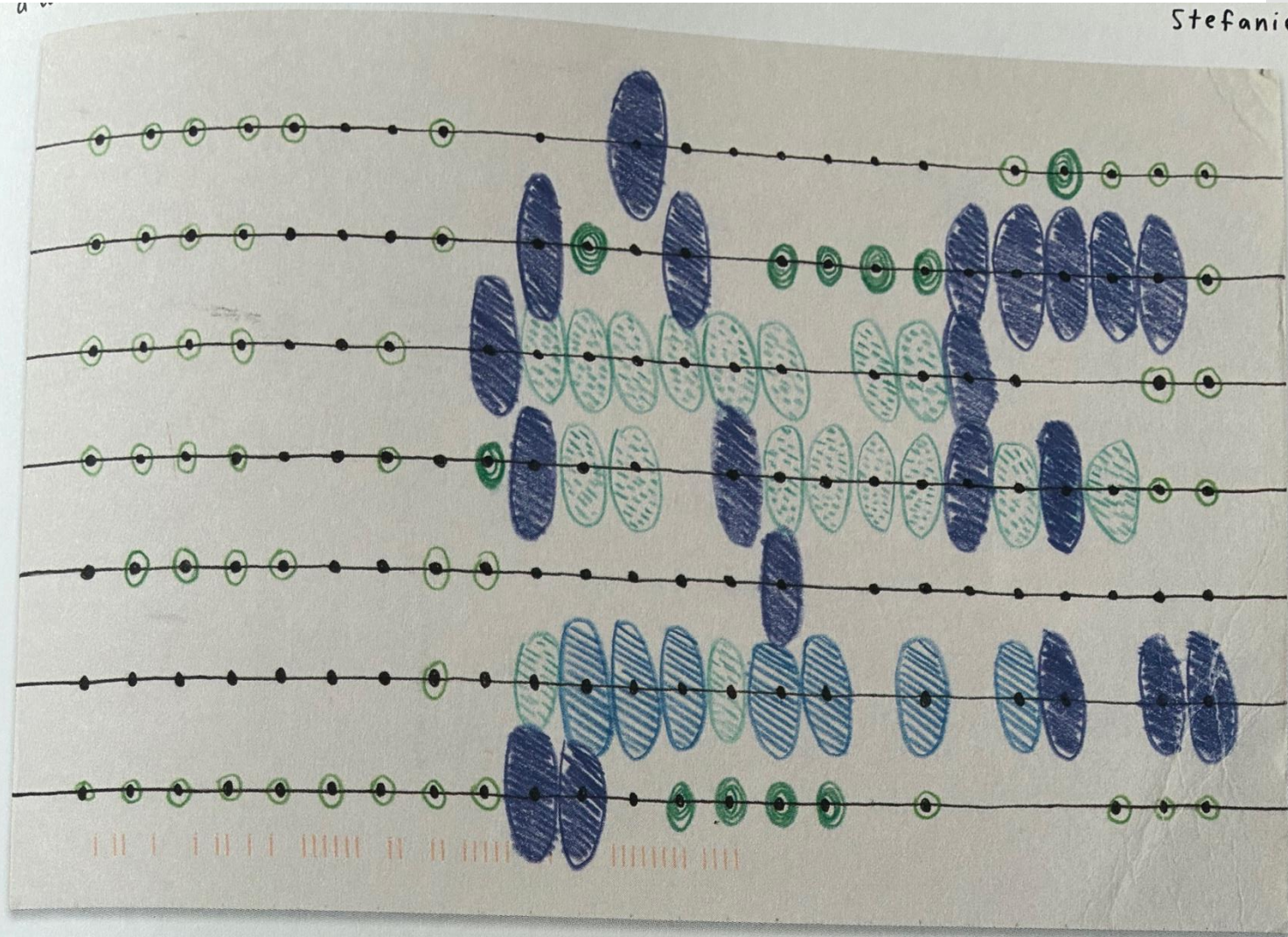


BLACK DOTS EACH REPRESENT AN HOUR WHERE THERE IS DATA.

### NUMBER OF PEOPLE IN MY SPACE/VICINITY:



Stefanie





# DEAR DATA - WEEK 24

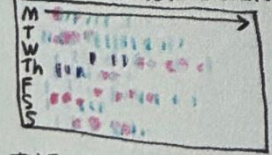
## A WEEK OF DOORS/SPACES

SECOND  
POSTING  
ARGH!!

ABOUT THE DATA: I GATHERED DATA ON ALL OF THE SPACES I PASSED THROUGH IN THE WEEK, BOTH INTERNAL + EXTERNAL. A SPACE IS DEFINED BY WHETHER I HAD TO PASS THROUGH A DOOR OR NOT.

### HOW TO READ IT:

IN CHRONOLOGICAL ORDER



EACH SPACE IS REPRESENTED BY A:

OR A (CROSS-HATCHED LINES  
CARRY NO IMPORTANCE)  
MAINLY ONLY FOR  
DIFFERENTIATION BETWEEN  
SIMILAR COLOURS! THIS

THE TYPES OF SPACES I PASSED THROUGH INCLUDE:

WAS A LAST MINUTE SPONTANEOUS  
DECISION ON THE 1ST CARD TO MAKE THE  
COLOURS CLEARER... SOOO ANNOYING I HAVE  
TO REDRAW !!

#### IN THE HOME/

##### PERSONAL SPACE:

+++++ BEDROOM  
SPARE BEDROOM  
(1 OF 2)  
GROUND FLOOR  
(OPEN PLAN, INCLS.  
KITCHEN + LOUNGE)  
TOILET  
BATHROOM

##### STUDIO:

+++++ STUDIO BLDG.  
STUDIO SPACE

##### OUTSIDE:

IN THE  
STREET

#### DOCTOR:

DOCTOR'S  
OFFICE  
WAITING  
ROOM

#### UNIVERSITIES:

UNIV. BLDG.  
CLASSROOM  
TEAROOM

#### CULTURAL CENTRE:

MAIN BLDG.  
GALLERY

#### TRANSPORT:

TUBE/TRAIN  
TUBE/TRAIN STN  
CAB

#### SHOP

BRYXTON MKT. BLDG.  
PUB/RESTAURANT/  
CAFE/BAR

#### FRIEND'S HOUSE:

MAIN HOUSE  
BASEMENT  
KITCHEN  
LOUNGE

#### GYM:

MAIN BLDG.  
CHANGING ROOM  
EXERCISE ROOM

\* CHANGING ROOMS +  
TOILETS ARE FOUND  
ACROSS DIFFERENT  
SPACES

FROM:

S. POSAVEC  
~~XXXXXXXXXX~~  
~~XXXXXXXXXX~~  
LONDON  
~~XXXXXXXXXX~~  
UK

To:



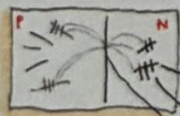
Stefanie



# DEAR DATA

## WEEK 29: MY BOYFRIEND

### HOW TO READ IT?



P = positive ones  
N = negative ones

This week I noticed and recorded everything that my boyfriend did that caused any emotional reaction on me.

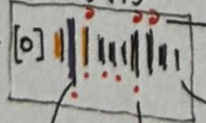
Each group represents a type of action  
Each single bar is the actual action, in chronological order

CONNECTIONS = things that can likely happen in a regular week = always

### [.] SYMBOL = TYPOLOGY

- P • helped me physically
- helped me in general
- \* took special care of me
- ♥ showed me love
- reassured me
- △ put my needs before his ones
- △ let me do weird stuff
- \* figured out stuff I couldn't
- showed me how wise and BETTER he is!

### ELEMENTS



red dot above = if present it means I said THANK you 😊

|| = "intensity of the action"

|| = regular || = very special THING

### COLOR = context

- work related
- Dear data rel.
- personal stuff
- life/practical stuff
- opinions
- objects/physical stuff
- the 2 of us

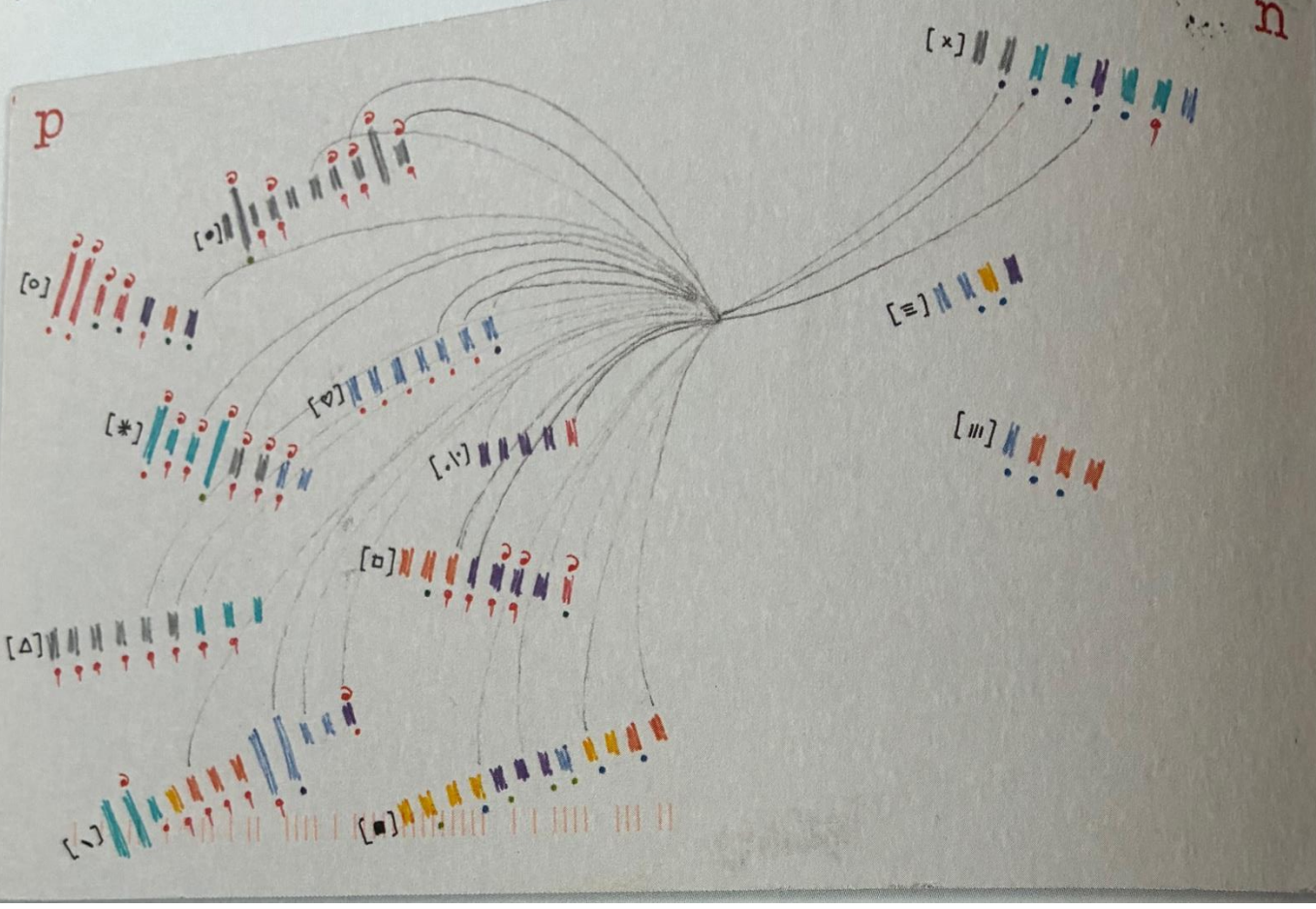
DOT BELOW = my reaction!

- surprised
- pleased
- felt better!
- annoyed
- imitated

void = nothing particular

wouldn't have noticed that for this Dear DATA week

GEORGIA



Georgia found this an incredibly partner trigger



## Your Turn

Find a small group to work with.

1. Pick a topic of interest and find data related to it.
2. Select two specific variables.
3. Sketch two different visual encodings of the variables. One should be simple, and one should be more creative/complex.